CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

JAK / TYK2: 2 molecules stopped early, all small molecules

4 stopped trials, but 2 molecules — all small molecules: Pacritinib and Ruxolitinib. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

5.9%
4 of 68 closed trials (95% CI 2.3%–14.2%)

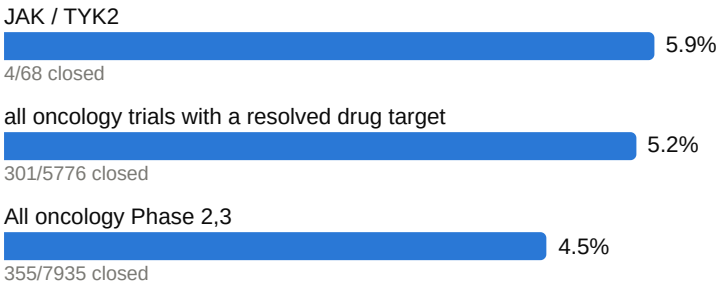
2 / 2 / 0
efficacy / safety / both (adds to 4)

49.3%
of 138 trials have closed · 70 still open or unresolved

1.9%
stopped for a biological reason within 3 years of starting (95% CI 0.5%–7.4%; 72 still at risk; 1.5% area-wide)

0 of 3 **Completed and missed the primary endpoint.** Of 44 completed trials in this segment, 3 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 0 missed and 3 met. Counted separately and never in the rate above; a floor, not a rate. 11 of the 40 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 135 distinct experimental drugs across 138 trials (small molecule 137, unknown 14, antibody 10, protein 7). Most tested: RUXOLITINIB (70), ITACITINIB (16), PACRITINIB (15), GOLIDOCITINIB (14), CYCLOPHOSPHAMIDE ANHYDROUS (11). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 4 stops came from 4 sponsor-asset programmes across 4 sponsors; the largest (Washington University School of Medicine) contributed 1; removing that programme's trials from both sides leaves 4.5%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 4 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 11.8%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	3	30	10.0%	3.5%–25.6%
2018–2020	0	20	0.0%	0.0%–16.1%
2021–2024	1	18	5.6%	1.0%–25.8%

Where the stops sit

Efficacy / futility: 2, Safety: 2.
Sponsors with most stops: Medical College of Wisconsin (1), M.D. Anderson Cancer Center (1), Incyte (1), Washington University School of Medicine (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04635059	2	Medical College of Wisconsin	PACRITINIB	2023-11	Efficacy	futility and drug supplier in favor of new metastatic protocol.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02420717	2	M.D. Anderson Cancer Center	CYCLOPHOSPHAMIDE ANHYDROUS, CYTARABINE, DASATINIB ANHYDROUS, DEXAMETHASONE	2021-01	Efficacy	Study was closed early due to low accrual and lack of response.
NCT02711137	2	Incyte INCY	INCB-057643, GEMCITABINE, PACLITAXEL, RUCAPARIB	2019-02	Safety	Study terminated due to safety issues.
NCT02277093	2	Washington University School of Medicine	PACRITINIB	2016-03	Safety	FDA issued a clinical hold as pacritinib had increased side effects

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 49.3% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.